

# European Union

## Rights Dimension (R1–R10)

Measures how strongly the framework protects individuals from AI harms. Max raw score: 20.

| Code | Indicator & Justification                                                                                                                                                                                                                                                                                                                                                                                                                                    | Score                             |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| R1   | <p><b>Binding comprehensive AI law with individual rights provisions</b></p> <p>Regulation 2024/1689 is a comprehensive binding regulation with individual rights provisions, mandatory conformity assessments for high-risk AI, and financial penalties up to €35M or 7% of global annual turnover. Entered into force August 1, 2024.</p>                                                                                                                  | Score 2 — Binding / comprehensive |
| R2   | <p><b>Explicit list of prohibited AI use categories in binding law</b></p> <p>Article 5 provides an explicit list of prohibited AI practices: social scoring by public authorities, real-time biometric identification in public spaces (with narrow exceptions), AI exploiting subconscious vulnerabilities, AI inferring emotions in workplace or education, biometric categorization revealing protected characteristics, and predictive policing AI.</p> | Score 2 — Binding / comprehensive |
| R3   | <p><b>Legal right to explanation for automated AI decisions</b></p> <p>Article 86 establishes a binding right to explanation: any affected person subject to a decision based on the output of a high-risk AI system has the right to obtain clear and meaningful explanations of the role of that system in the decision-making procedure.</p>                                                                                                              | Score 2 — Binding / comprehensive |
| R4   | <p><b>Legal right to human review of AI decisions</b></p> <p>Article 14 requires that all high-risk AI systems be designed and developed with human oversight measures enabling natural persons to effectively oversee the system and to intervene or interrupt its operation when necessary.</p>                                                                                                                                                            | Score 2 — Binding / comprehensive |
| R5   | <p><b>Legal right to contest or appeal AI-driven decisions</b></p> <p>Recital 175 and Article 86 together establish a right to contest automated decisions. Deployers must inform affected persons of their right to lodge complaints. Article 89 provides for individual rights of redress.</p>                                                                                                                                                             | Score 2 — Binding / comprehensive |

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| <b>R6</b>  | <b>Anti-discrimination provisions specifically targeting AI</b> <p>Article 10 requires high-risk AI training data to be examined for possible biases. Article 5(1)(c) prohibits AI that exploits characteristics or vulnerabilities to distort behaviour. Recital 45 explicitly addresses AI systems that may discriminate based on protected characteristics.</p>     | <b>Score 2 — Binding / comprehensive</b> |
| <b>R7</b>  | <b>Independent enforcement body acting on behalf of individuals</b> <p>The EU AI Office was established in 2024 as the central enforcement body for GPAI models. Each of the 27 member states must designate national market surveillance authorities. Article 74 provides for coordinated enforcement.</p>                                                            | <b>Score 2 — Binding / comprehensive</b> |
| <b>R8</b>  | <b>Meaningful financial penalties calibrated to harm or global turnover</b> <p>Article 99 establishes penalties: up to €35,000,000 or 7% of total worldwide annual turnover for prohibited AI violations; up to €15,000,000 or 3% for other obligations; up to €7,500,000 or 1.5% for supplying incorrect information. Penalties are calibrated to global revenue.</p> | <b>Score 2 — Binding / comprehensive</b> |
| <b>R9</b>  | <b>Restrictions on biometric surveillance or invasive AI in binding law</b> <p>Article 5(1)(d) prohibits real-time remote biometric identification in publicly accessible spaces with narrow law enforcement exceptions subject to judicial authorisation, time limits, and geographic constraints.</p>                                                                | <b>Score 2 — Binding / comprehensive</b> |
| <b>R10</b> | <b>Constitutional or treaty-level AI rights protection</b> <p>The EU AI Act is a regulation, not a constitutional instrument. There is no specific constitutional protection for AI rights at EU level. Chile's neuro-rights constitutional protection (2021) is the only example globally of Score 2 on this indicator.</p>                                           | <b>Score 0 — Absent</b>                  |

## Market Dimension (M1–M10)

*Measures how strongly governance relies on voluntary, industry-led mechanisms. Max raw score: 20.*

| Code | Indicator & Justification | Score |
|------|---------------------------|-------|
|------|---------------------------|-------|

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| <b>M1</b> | <b>Primary governance mechanism is voluntary, not legally binding</b><br><p>The EU AI Act is primarily binding legislation with mandatory conformity assessments, documentation requirements, and financial penalties. Score 0.</p>                                                                           | <b>Score 0 — Absent</b>                           |
| <b>M2</b> | <b>Industry self-governance is the primary compliance mechanism</b><br><p>Compliance is driven by state enforcement through national market surveillance authorities and the EU AI Office. Score 0.</p>                                                                                                       | <b>Score 0 — Absent</b>                           |
| <b>M3</b> | <b>Innovation promotion explicitly mandated in governance framework</b><br><p>The Act includes regulatory sandboxes (Articles 57–63) and exemptions for research activities. Recital 2 explicitly states the Act aims to support innovation. Score 1 — innovation as secondary goal, not primary mandate.</p> | <b>Score 1 — Soft / partial / sector-specific</b> |
| <b>M4</b> | <b>No binding financial penalties exist for AI violations</b><br><p>The Act imposes substantial penalties up to €35M or 7% global turnover. Score 0.</p>                                                                                                                                                      | <b>Score 0 — Absent</b>                           |
| <b>M5</b> | <b>Formal regulatory sandbox or experimental zone for AI</b><br><p>Articles 57–63 require all member states to establish national AI regulatory sandboxes by August 2026.</p>                                                                                                                                 | <b>Score 2 — Binding / comprehensive</b>          |
| <b>M6</b> | <b>Framework is principles-based rather than prescriptive-rules-based</b><br><p>The Act is highly prescriptive for high-risk AI: mandatory conformity assessments, technical documentation, transparency obligations, logging requirements, accuracy and robustness standards. Score 0.</p>                   | <b>Score 0 — Absent</b>                           |
| <b>M7</b> | <b>No dedicated enforcement body with AI sanctioning powers</b><br><p>The EU AI Office plus 27 national market surveillance authorities and 27 national competent authorities. Score 0.</p>                                                                                                                   | <b>Score 0 — Absent</b>                           |
| <b>M8</b> | <b>Compliance is primarily self-assessed, not externally audited</b><br><p>High-risk AI requires external conformity assessment (Article 43) or third-party audit by notified bodies. Score 0.</p>                                                                                                            | <b>Score 0 — Absent</b>                           |

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| <b>M9</b>  | <b>Explicit alignment with international voluntary standards (OECD, ISO 42001)</b><br><br>Recital 168 explicitly aligns the Act with OECD AI Principles. The EU co-authored or influenced OECD, ISO, and IEEE AI governance standards. Score 2. | <b>Score 2 — Binding / comprehensive</b> |
| <b>M10</b> | <b>Light-touch, pro-industry approach explicitly stated in official policy</b><br><br>The Act is explicitly protective and regulatory. No official EU document describes the AI Act as light-touch. Score 0.                                    | <b>Score 0 — Absent</b>                  |

## Sovereign Dimension (S1–S10)

*Measures how strongly the state directs and controls AI development and deployment. Max raw score: 20.*

| Code      | Indicator & Justification                                                                                                                                                                                                                   | Score                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>S1</b> | <b>AI outputs must comply with state values or ideology in binding law</b><br><br>No requirement in the EU AI Act for AI outputs to comply with state values or ideology. Score 0.                                                          | <b>Score 0 — Absent</b>                           |
| <b>S2</b> | <b>Mandatory state registration required before AI deployment</b><br><br>No mandatory state registration before AI deployment. The EU database (Article 71) is a public transparency measure, not a state permission requirement. Score 0.  | <b>Score 0 — Absent</b>                           |
| <b>S3</b> | <b>National security provisions give state broad AI override powers</b><br><br>Article 2(3) explicitly excludes AI systems used exclusively for national security from the Act's scope. This is a carve-out, not a state override. Score 1. | <b>Score 1 — Soft / partial / sector-specific</b> |
| <b>S4</b> | <b>State directs AI development priorities through binding strategic mandate</b><br><br>The EU AI Act does not direct AI development toward specific strategic economic goals. Score 0.                                                     | <b>Score 0 — Absent</b>                           |
| <b>S5</b> | <b>Data localization requirements affecting AI training or deployment</b><br>No data localization requirements in the EU AI Act. Score 0.                                                                                                   | <b>Score 0 — Absent</b>                           |

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| <b>S6</b>  | <b>Restrictions on foreign AI providers or models in binding law</b><br>No restrictions on foreign AI providers. The Act applies based on where AI is placed on the market. Score 0.                       | <b>Score 0 — Absent</b> |
| <b>S7</b>  | <b>State ownership or investment explicitly directs AI development</b><br>No state ownership requirements. Score 0.                                                                                        | <b>Score 0 — Absent</b> |
| <b>S8</b>  | <b>State surveillance AI explicitly authorized or mandated in law</b><br>State surveillance AI is restricted, not authorized. Article 5 prohibits most real-time biometric surveillance. Score 0.          | <b>Score 0 — Absent</b> |
| <b>S9</b>  | <b>Content moderation requirements serving state political interests</b><br>No content moderation requirements serving political interests. The Act prohibits manipulation of political opinions. Score 0. | <b>Score 0 — Absent</b> |
| <b>S10</b> | <b>State pre-approval required for AI deployment</b><br>No state pre-approval required. Market surveillance is post-deployment monitoring. Score 0.                                                        | <b>Score 0 — Absent</b> |

### Derived Coordinates

Ternary coordinates (r, m, s) are computed directly from the raw indicator sums. The compass rx coordinate is derived as  $(r - s) \times 1.15$ , capped at  $[-1, +1]$ . Bind score reflects enforcement strength independently of the ternary position.

|                      |           |    |                         |
|----------------------|-----------|----|-------------------------|
| <b>Rights (R)</b>    | <b>18</b> | 20 | <b>r = 0.75</b>         |
| <b>Market (M)</b>    | <b>5</b>  | 20 | <b>m = 0.21</b>         |
| <b>Sovereign (S)</b> | <b>1</b>  | 20 | <b>s = 0.04</b>         |
| <b>Total</b>         | <b>24</b> | 60 | Enforcement (bind): 95% |

$$r = 18 \div 24 = 0.75 \quad m = 5 \div 24 = 0.21 \quad s = 1.00 - 0.75 - 0.21 = 0.04$$